

Math 514, F13
Quiz 9

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) The current price of a security is 40. Given a nominal interest rate of 6%, compounded continuously, find a lower bound for the price of a call option that expires in six months with a strike price of 38.

put-call option parity formula.

$$S + P - C = Ke^{-rt} \rightarrow (P \geq 0, C \geq 0)$$
$$C = S + P - Ke^{-rt}$$
$$\Rightarrow C \geq S - Ke^{-rt} \text{ since } P \geq 0$$
$$= 40 - 38e^{-0.06 \times (\frac{1}{2})}$$
$$\approx \underline{3.123}$$

Problem 2. (6 points) An European call and put option on the same security both have the expiration time $t = 2$ years and the strike price $K = 25$. The call option sells for 3 (i.e. $C(25, 2) = 3$) and the put option for 6 (i.e. $P(25, 2) = 6$). Assume the nominal continuously compounded interest rate is 10% and the current stock is 20. Identify an arbitrage.

$$S + P - C = 20 + 6 - 3 = 23$$
$$Ke^{-rt} = 25 \cdot e^{-0.1 \times 2} = 20.468$$
$$\Rightarrow S + P - C \geq Ke^{-rt}$$

An arbitrage can be obtained by selling the security short, selling a put option and buying a call option:

Initial yield: $20 + 6 - 3 = 23$

Cost at $t = 2$: 25

The gain (present value) is thus $23 - 25e^{-0.1 \times 2}$

$$= 23 - 20.468$$
$$= 2.532$$